

TECHNICAL ARTICLES

INSIGHTS, INNOVATIONS
AND IN-DEPTH KNOWLEDGE
FROM EXPERTS

EXPERT PERSPECTIVES

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TECHNOLOGY
IS BEST WHEN
IT BRINGS PEOPLE
TOGETHER.



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From the Desk of President

Dear CSE Department,

It is with great pleasure that I extend my heartfelt greetings to the talented members of our Computer Science and Engineering family. As we embark on another academic journey, I am filled with pride and enthusiasm for the remarkable achievements and unwavering dedication that define our CSE department.

In the ever-evolving landscape of technology, our CSE department continues to be a beacon of excellence. Our students, faculty, and staff contribute tirelessly to the dynamic field of computer science, pushing boundaries and shaping the future.

The collective brilliance within these walls is not only inspiring but also a testament to our commitment to fostering innovation, collaboration, and a relentless pursuit of knowledge.

I commend the initiative to showcase the department's achievements through this magazine, "Prisma 3.0". Let the pages within be a reflection of our shared journey, celebrating successes, acknowledging challenges, and highlighting the collective spirit that defines us. I encourage everyone to actively participate and contribute to making this edition a true representation of the outstanding work emanating from our CSE family.

Wishing you all a year filled with intellectual growth, innovative endeavors, and collaborative triumphs.

Best Regards,
Dr. Sourabh Kumar Das
President,
Prisma

Kalyani Government Engineering College

From the Desk of Secretary

Technology can be a powerful medium to transform our world, but its true significance emerges through curiosity, empathy, and creativity. As followers and learners of Computer Science, logic forms the foundation of our thinking, but the true impact of our innovations also depends on our character and human values.

Beyond algorithms and code, we should strive to develop the art of storytelling, adaptability, and ethical leadership in every step of life. Outside the classroom, PRISMA 3.0 serves as a wonderful platform to showcase our artistic talents, unique perspectives, and creativity.

May your code remain clean, but your thoughts be far-reaching, and may your curiosity and passion for learning something new never fade.

At this point I also want to thank all the esteemed Alumni of the department, for their help and contribution at different times in our journey.

My sincere thanks to The Principal, all H.O.Ds, The Registrar and the whole KGEC Family without their help this would not have been possible.

My greeting to the editorial board, students, faculty, teacher, and staff members of the CSE department to keep on doing the good work for the students, department, and society.

I hope all of you enjoy the Prisma 3.0

Best Regards,
Dr. Kousik Dasgupta,
Secretary,
Prisma

Kalyani Government Engineering College



STUDENTS' MAGAZINE COMMITTEE





The War on Hallucinations

Beyond Stochastic Parrots: Hierarchical Verification and Self-Correction in LLMs

Introduction: The Mathematical Inevitability of Hallucinations

Large Language Models (LLMs) have irrevocably transitioned from experimental prototypes into mission-critical infrastructure. As these intelligent systems are deployed in high-stakes environments—ranging from automated software engineering to autonomous financial execution—the engineering mandate has shifted from scaling parameter counts to enforcing probabilistic reliability.

What Is an AI Hallucination?

In Plain English: An AI hallucination happens when an AI model confidently makes something up. It doesn't say "I don't know"—it presents fabricated facts, invented statistics, fake legal cases, or non-existent medical studies as if they were real. The response sounds authoritative and reads perfectly. That is exactly what makes it so dangerous.

Technical Definition: Hallucination refers to generated output that is not grounded in the provided input data or factual reality. There are two primary types:

Intrinsic hallucination (faithfulness hallucination): The model contradicts information explicitly provided in its source material.

For example, during summarisation, it adds facts not present in the original document.

Extrinsic hallucination (factuality hallucination):

The model generates information that cannot be verified against any known source—it invents facts, citations, statistics, or events from scratch.

Why It Happens?

Large Language Models (LLMs) are fundamentally prediction engines, not knowledge bases. They generate text by predicting the most statistically likely next word based on patterns learned from training data. They do not understand truth, they predict plausibility. When the model encounters a gap in its training data or faces an ambiguous query, the mathematics of its softmax function force it to guess the most plausible-sounding continuation, bridging gaps in logic with seamless fiction. For computer science engineers, the mandate is clear: we can no longer treat these hallucinations as traditional software "bugs" that can be patched. Because this stochastic behaviour is mathematically inevitable, absolute reliability must be engineered through external architectural oversight.

Engineering Reliability:

Reflection and Hierarchical Verification

Contemporary research has moved decisively away from single-pass inference towards multi-layered verification architectures. Instead of relying on a single, flawed generative pass, we deploy both internal and external auditing mechanisms to force the model to reason accurately.

Intrinsic Reflection (Self-Correction)

The first line of defence is intrinsic reflection. Instead of outputting its initial statistical prediction, the model is forced into an iterative self-critique loop. It drafts a response in a hidden scratchpad, and a secondary system prompt instructs the model to critique its own logic checking for contradictions and unsupported claims before delivering the final output.



Extrinsic Verification (Multi-Model Orchestration):

However, a model cannot always detect its own blind spots. When intrinsic reflection fails, the state-of-the-art relies on extrinsic hierarchical verification, known as the Proposer-Verifier-Synthesiser pattern. This mirrors dual-process theory in cognitive psychology by dividing tasks into intuitive generation and analytical reasoning.

Proposer : Generates a high-entropy draft, exploring the latent space with deliberate creative latitude. It prioritises recall and possibility exploration over strict factual adherence.

Verifier : A deterministic, specialised auditing layer. Frameworks like the Verifiable Iterative Refinement Framework (VIRF) utilise formal logic (such as Web Ontology Language or OWL rules) to parse and evaluate proposed actions.

Synthesiser: Integrates only the verified claims into a final output, ensuring the response is logically coherent and explicitly auditable.

The Power of Pedagogical Feedback : Crucially, these systems work together dynamically. Modern Verifiers don't merely reject incorrect outputs; they generate pedagogical feedback. For example, if the Proposer suggests using an $O(n^2)$ sorting algorithm for a massive dataset, the Verifier doesn't just return "Error." It outputs a logical proof: "Constraint violation: $O(n^2)$ complexity on large data causes timeout. Memory limit exceeded. Suggestion: Implement external merge sort." This feedback is fed directly back into the Proposer's intrinsic reflection loop, allowing the system to self-correct and generate a mathematically sound solution.

Empirical benchmarks highlight the success of this approach. In tests mirroring Anthropic's Constitutional AI, injecting a Verifier reduced hallucinated citations by over 50% while maintaining high recall, proving that separating generation from validation is fundamentally superior to single-pass prompting.

RAG 2.0 : Graph-Based Contextual Anchoring

While reflection fixes logical errors, factual retrieval requires structural anchoring. Traditional Retrieval-Augmented Generation (RAG) struggles with "semantic drift" across complex queries.

GraphRAG addresses this by representing knowledge as structured Knowledge Graphs rather than disconnected vector embeddings. Using advanced clustering algorithms, GraphRAG allows models to traverse explicit relational edges, creating an auditable reasoning path. While GraphRAG yields a massive 50–70% improvement in global summarisation and excels at multi-hop reasoning, it introduces significant trade-offs compared to traditional vector RAG: high index construction costs, a 2.3× increase in query latency, and up to a 13.4% accuracy degradation on highly specific factual queries. Furthermore, GraphRAG is vulnerable to poor metadata governance; noisy source data creates convoluted graph topologies. To fix this, engineers are adopting programmatic optimisation frameworks like DSPy (Declarative Self-Improving Python). Instead of relying on brittle manual prompts, DSPy algorithmically tunes smaller models to perform strictly typed, schema-compliant entity extraction. This ensures only semantically valid relationships enter the knowledge graph, which drastically reduce hallucination rates.

The Future: Reason-to-Code Architectures.

To achieve zero-defect reliability in algorithmic domains, systems engineers have formalised the Reason-to-Code loop. Acknowledging that neural networks are fundamentally unsuited for precise arithmetic due to floating-point attention mechanism



the model is forced to output formal code rather than a direct numerical answer.

This code executes in a secure, sandboxed environment. The interpreter acts as a deterministic oracle, feeding the verified computational result back to the model for final natural language synthesis. By bypassing probabilistic attention heads in favour of deterministic CPU execution, we neutralise the mathematical inevitability of hallucination.

Conclusion

Recent theoretical work reframes the "stochastic parrot" critique by defining semantic grounding not as an internal hidden state, but as a verifiable predictive relationship between symbols and observable phenomena.

This perspective aligns perfectly with modern architecture. We are no longer treating hallucination mitigation as a superficial patch. Through intrinsic reflection, hierarchical verification, DSPy-optimised GraphRAG, and Reason-to-Code loops, we are fundamentally evolving LLMs from probabilistic text generators into verifiable cognitive engines. For the systems architects of tomorrow, this represents the true frontier.





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AI and the Akasha Terminals

Sumeru's Dream of Knowledge

In Sumeru, knowledge was everywhere. It did not live only in books, classrooms, or the halls of the Akademiya. It rested lightly against the ears of the people in the form of the Akasha Terminal, a small device that connected every citizen to a vast network of information. With it, facts could be retrieved in an instant, questions could be answered almost before they were fully formed, and the pursuit of wisdom seemed easier than ever. Sumeru, the nation devoted to knowledge, appeared to have achieved a scholar's dream: a society where information flowed freely through every mind.

But beneath that miracle was a quiet loss.

The Akasha was originally created from the power of the Dendro Archon, Greater Lord Rukkhadevata, as a system to help guide and support her people. After her death in the cataclysm, Lesser Lord Kusanali — Nahida — inherited a nation that praised wisdom while ignoring its own god. The sages of the Akademiya, dissatisfied with Nahida and blinded by their own pride, came to believe that Sumeru needed a new deity: a God of Wisdom shaped not by nature or divinity, but by human design.

Their answer was the Akasha.

Using the network that connected all of Sumeru, the sages carried out a grand and disturbing experiment. They harvested dreams from the people, especially during the Sabzeruz Festival, trapping the citizens of Sumeru City in a samsara so they could gather enormous amounts of data from their repeated dreams. This was not simply a machine giving answers. It was a machine built on the minds of the people themselves. The sages then used the Akasha's stored knowledge along with the Electro Gnosis to create Scaramouche's mechanical god-body and elevate him toward artificial divinity.

At the center of all this was Nahida, imprisoned in the Sanctuary of Surasthana, dismissed as powerless by the very scholars who claimed to protect wisdom. Yet Nahida saw what the sages could not. Knowledge without compassion becomes arrogance. Information without understanding becomes emptiness. And a nation that worships answers too much may forget how to ask the right questions.

When the conflict reached its climax, Nahida did not simply destroy a machine and declare victory. She confronted Scaramouche with patience, intelligence, and extraordinary resolve. Through repeated battles and careful preparation, she gathered the means to defeat him, exposing the emptiness behind the sages' ambition. Their dream of manufacturing a god collapsed. In the aftermath, Nahida took control of the future of Sumeru for herself.

One of her most important choices came after the battle. She shut down the Akasha system. This decision mattered because the Akasha had become more than a tool. It had become a crutch. For generations, the people of Sumeru had grown used to receiving knowledge instantly, depending on the system instead of exercising their own memory, judgment, and curiosity. The nation that loved wisdom had, in some ways, become distant from the struggle that true understanding requires.

That is what makes the Akasha feel so close to our own world.

Today, people turn to AI in much the same way the people of Sumeru turned to their terminals. They ask it to explain, summarize, write, organize, and decide. AI can be helpful, just as the Akasha was helpful. It can widen access to knowledge, save time, and reduce effort. But the danger begins when help becomes dependence.



The more people rely on a system to remember for them, think for them, or create for them, the easier it becomes to let their own abilities grow dull.

The Akasha was never evil by nature. Its tragedy came from the way it reshaped the habits of an entire society. It made knowledge immediate, but perhaps too easy. It connected everyone, but also made them vulnerable to control by those who ruled the system. In the wrong hands, a network meant to share wisdom became a structure for surveillance, manipulation, and ambition.

AI carries a similar tension. It is not a god, and it is not the Akasha, but it raises the same question: what happens when humans stop treating intelligence as something to cultivate, and begin treating it as a service always available on demand? Convenience is seductive. Efficiency is persuasive. And dependence rarely announces itself loudly. It arrives quietly, through habit. Nahida's story offers a gentler answer than simple fear. She does not reject knowledge, and she does not reject innovation. Instead, she restores balance. She frees Sumeru from a system that had begun to think in place of its people. In doing so, she reminds us that wisdom is not measured by how quickly answers appear, but by how deeply people understand them.

The fall of the Akasha was not the end of knowledge in Sumeru. It was the beginning of something more human. People would have to remember, study, question, and learn again without leaning so heavily on the voice in their ears. That change was not as effortless as the old system had been, but it was freer.

And perhaps that is the real lesson for us now. AI can illuminate, assist, and inspire. But if we let it become our Akasha Terminal — our constant, unquestioned source of thought — we may gain speed while losing something quieter and more precious: the habit of thinking for ourselves.



If AI Existed In 19th Century

Science Fiction Or Scientific Possibility?

This is the 21st century and it is the era of AI, from studies to business and offices AI is everywhere, people are undoubtedly benefiting a lot from this because this complex system has made their lives very much easier, but think about existence of AI two centuries ago,, there was no AI, the 19th century was of steam engines, industrial revolutions and empires. What if these two eras collide? I mean what if AI existed in the late 19s century

Today AI is transforming civilizations, transforming our way of living. Think about education, healthcare , robotics, transportation , entertainment ,cybersecurity , and what not?. AI has made everything easier. But in 19th century or even before that why was AI not invented?. actually AI heavily depends on modern computation, huge amounts of data and algorithms. The 19th century lacked modern computers , it was not the lack of data but the storage was unavailable, data was only stored in pen and paper. And the level of algorithms needed to build AI was also not available at that time.

Yet imagining a scenario where AI existed in 19th century reveals both the power and limitation of AI.

AI is largely based on data and algorithms. The main part on which AI heavily depends is machine learning. Developers have to train the computer through huge amounts of data called datasets, they train their programs in such a way that they can easily recognise patterns and tackle complex problems. The algorithm works through various layers called neural networks. These neural networks extract patterns from examples in training datasets. They search for patterns in multiple steps. Once a programme has been trained on a dataset, it can then encounter new scenarios and also can make predictions.

All these happen after the programme is trained by various examples by the dataset.

Now if we imagine these things in the 19th century, this sounds absurd, maybe the words did not exist like networks or datasets. But if we think deeply a non developed AI system could have existed because at that time huge amounts of data were not stored there but some data were still there which could have been used to predict outcome. Think of it like a baby, his parents teach him when to do what,, and once after some years the kid takes decisions of his own,, this is his own intelligence. A similar thing happens in AI but artificially.

So why not in the 19th century?.it could have existed but like a primitive child because AI needs many things which were not there in the past,, like no GPUs, no transistors , very limited storage and above all no internet, the machines were mostly mechanical, computation was still there in that time but at a very primary level. One example is analytical engine invented by Charles Babbage in 1837. So as said previously, AI could have existed like a primitive child, although modern AI would be impossible , primitive mechanical intelligence or rule based systems might still have emerged.

Now lets discuss about how AI could have exist at that time and about some historical events that could have changed, not taking too much things, I am here trying to discuss about some things which are A. Industrial Revolution B. Medicine and Epidemics C. Wars and Colonialism D. Scientific Discoveries.

Industries and factories could have been greatly affected by AI. Several things could have changed like factories can use smart ways to produce products with less manpower and less expenses.



AI could have predicted ways to reduce cost and to improve product quality. In old systems maintenance cost was also too high. Machines often broke down due to several reasons. AI could have prevented these things if it existed. Sometimes companies feared to take risks for producing huge amounts of products. AI could have predicted what can happen if the risk was taken, so that companies could invest to produce large amounts of products calculating the probability. Production would also become faster with the help of AI. But as works get easier, companies would need fewer people for work, so unemployment would be higher because of these, which is now visible in 21st century. People are facing huge number of layoffs in companies. If AI was there in 19th century improvement would happen earlier but this modern automation fear would also have been started earlier. Now coming to medicine and healthcare. AI could have predicted the possibility of outbreaks of disease earlier. Many people could have survived pandemics and epidemics if people had known about them earlier. Even it could have suggested preventive measures of that pandemic or epidemic or ways to defeat them. Better ways of sanitization would be suggested by AI. So that the outbreak of deadly diseases in the 19th and 20th century like plague and dengue might not have happened. Because of this mortality rates would be reduced significantly. Today AI is used in detecting diseases from x-rays, MRI scans, and reports, predicting health risk and patient condition, discovering and testing new medicines, assisting in robotic surgeries and in many fields. Maybe AI could not do so many things in those early days but even if there was a primitive form of AI in those days, the medical field would have been much more developed today.

AI could have had a great impact on the historical wars and colonialism that we read about in history books today.

Maybe the history would have changed if AI existed. AI-assisted military systems and strategies could have changed the history. It could help detect or predict enemy plans, analysing possible outcomes of steps taken. Besides these many things like tracking enemy camps and immediately sending alerts if any movement was noticed (auto surveillance system) could have emerged on those days. Faster communication in the time of needs could have existed. Many propaganda on those days were created to provoke normal persons by foreign powers and even by the rulers on the subjects. Which had affected history very much. AI could analyse those things and the true intentions behind them. Because of these the entire history of colonialism would have been changed.

Scientific discoveries would also be helped by AI in various ways. Even science would be much more developed today if AI existed earlier. AI could have helped scientists to recognise patterns, predict outcomes of certain experiments, make calculations easier. Often in past scientists overlook a thing which really mattered for that experiment, AI could warn them about such things. In many cases they have to wait for a very long time to get the outcome. AI could have predicted the outcomes earlier by prediction, early pattern analysis and some complex mathematical problems. If great scientists like Nikola Tesla, Charles Darwin, Gregor Mendel and many more had received the benefit of AI at that time, they could publish their research earlier. Which would open countless new possibilities to make new discoveries.

Still there were many fields where AI could make an impact. But every coin has two sides. If AI was discovered earlier society would be greatly benefited by AI without any doubt but still some questions remain. Would people depend on machines earlier?.. would empires misuse AI?



would it create inequality among people? Because everyone could have not the access to these.. would workers lose their jobs faster?.. we are getting some answers in our era. Yes, some people are getting too dependent on AI. Some are being too lazy because of this. Some people have started misusing AI. People are losing their jobs too. But after all what is the conclusion?. Is AI bad for us?. Genuinely I think we can't answer these questions, maybe in future we may find the answers , but if AI had existed earlier and if we have seen the consequences of it in past years, this question would be answerable.

Artificial intelligence is not merely a technology. It is a product of centuries of mathematics, engineering, physics and above all, the human curiosity. Imagining AI in the 19th century not only reshapes history in our minds but also helps us understand the immense impact of AI in our future.



Linux Kernel Development

Where The Software Ends

The Invisible Engine

Every time we open a browser, save a file, or stream a video, a silent conductor orchestrates billions of hardware events per second. That conductor is the Linux kernel. For decades, it has been dismissed as mere “plumbing” — complex, arcane, best left to grey-bearded wizards. But look closer, and we will find one of the most vibrant, creative, and welcoming open-source communities on the planet. Far from being a fossil, the Linux kernel is a playground of bold ideas, from running miniature sandboxed programs inside the kernel to rewriting critical subsystems in a memory-safe language. This essay takes us on a tour of that innovation — not as a textbook, but as a story.

A Culture of Continuous Reinvention

The kernel’s development model is itself an innovation. Over two thousand developers contribute to each release, following a strict but transparent process. Anybody can read the Linux Kernel Mailing List (LKML), and anybody can propose a patch. What keeps the kernel from collapsing into chaos? A combination of Linus Torvalds’ final say, lieutenants for each subsystem, and an unwavering rule: never break user space. This stability paradox — constant change without breaking existing programs — is the kernel’s superpower.

One of the most exciting recent inventions is eBPF. Originally designed for packet filtering, eBPF now lets us attach tiny, verified programs to almost any kernel event: system calls, function entry and exit, network packets, even tracepoints. These programs run in a sandbox, so they cannot crash the kernel. This has revolutionised observability.

Tools like ‘bpftool’ allows us to write one-liners that peek inside the scheduler or file system without recompiling anything. Imagine asking “How long does ‘open()’ take for my database?” and getting a histogram of latencies within seconds. That is eBPF.

Another quiet revolution is the introduction of Rust into the kernel. Starting with Linux 6.1, developers can write drivers and modules in Rust, a language that guarantees memory safety at compile time. No use-after-free, no data races. This does not replace C — the kernel is far too large for that — but it opens a safe path for new code. Writing a driver for a quirky sensor or a new filesystem becomes less about chasing pointer bugs and more about logic. The Rust for Linux project is a beautiful example of how a conservative project can embrace radical ideas.

Where the Action Is Today

If we want to see where kernel developers are spending their weekends, look at scheduling and memory management. The Linux scheduler (currently EEVDF) constantly evolves to balance responsiveness and throughput on machines with hundreds of cores. Meanwhile, memory management researchers are tackling the arrival of disaggregated memory — remote memory pools connected by CXL or networking.



The kernel's page reclaim and swapping algorithms were designed for local DRAM; making them work efficiently with remote, slower memory is an open engineering challenge that will shape the data centres of tomorrow.

Real-time Linux has also matured. With the PREEMPT_RT patches mostly merged, the kernel can now guarantee response times in microseconds. This is why we find Linux inside robot arms, medical devices, and audio workstations. The innovation here is not a single feature but a sustained, decade-long effort to reduce latency everywhere.

Resources to Start A Journey

We do not need to be a kernel maintainer to play. The official source tree (github.com/torvalds/linux) is intimidating but rewarding. Start with the 'Documentation/' folder — it is surprisingly readable. For hands-on experiments, the Linux Kernel Module Cheat repository gives us a QEMU setup where we can crash and rebuild the kernel in seconds. And if eBPF excites you, the BCC tools repository contains dozens of ready-to-run scripts that show what eBPF can do. Finally, the Kernel Newbies website and the #kernelnewbies IRC channel on OFTC are famously friendly places to ask "stupid" questions (none are stupid).

A Few Open Horizons

Innovation in the kernel rarely comes from grand plans. Often it emerges from someone scratching an itch.

For instance, could we use eBPF to let user-space programs ask the kernel "Which CPU should I run on next?" and get a real-time answer based on cache misses? Could we build a live-patching tool that uses Rust's safety guarantees to hot-swap functions without risking a crash? Another idea: a visual scheduler explorer that turns 'perf sched' traces into an animated movie of threads bouncing between cores — a tool that would delight students and professionals alike. These are not research projects; they are weekend challenges.

Why It Matters

The Linux kernel is often called "the largest collaborative project in human history." But that label misses the point. It is not large because it has to be — it is large because it works. Every innovation, from eBPF to Rust to real-time scheduling, expands what we can build on top of it. And the best part? The door is open. we do not need permission. Clone the source, write a tiny patch that prints "hello" when someone plugs in a USB drive, send it to the mailing list, and see what happens. The kernel is not a museum — it is a conversation.

For further reading: kernel.org/doc, ebpf.io.



